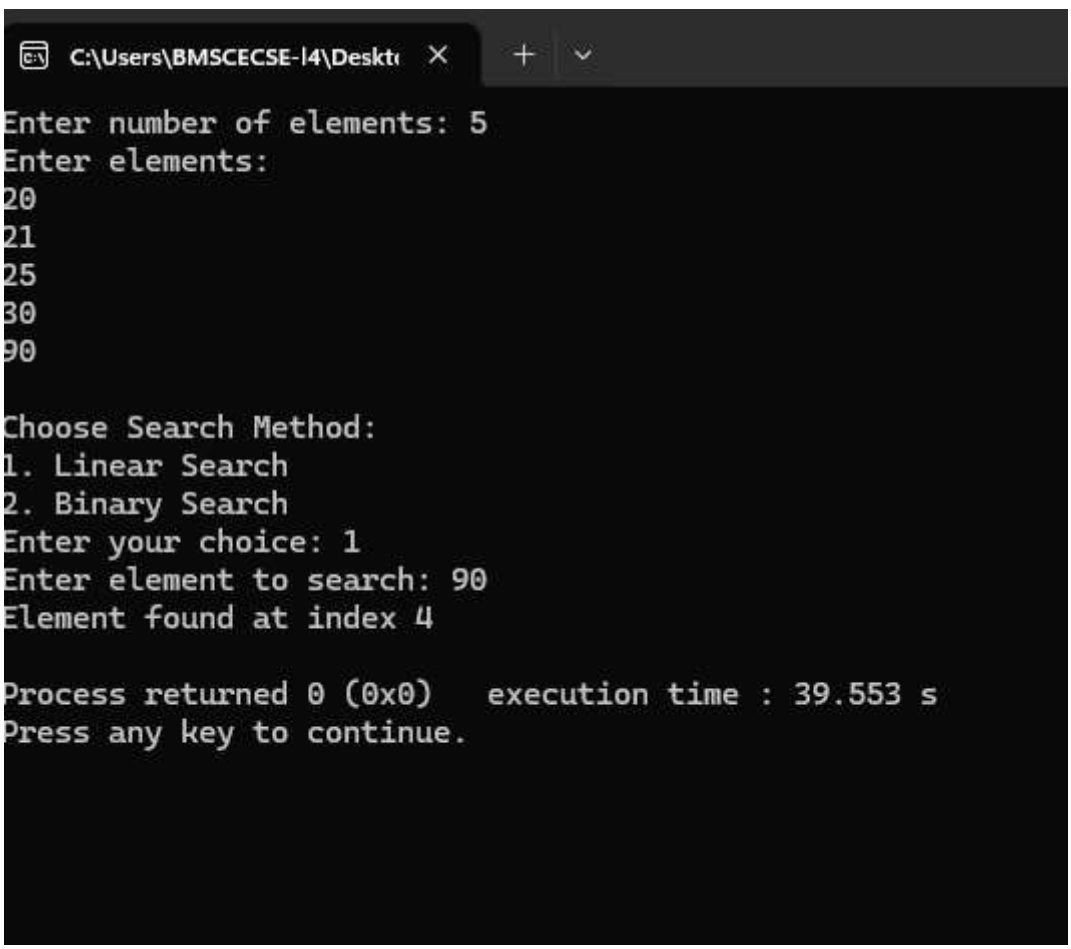
1.Write a C program to perform the following a) Linear search b) Binary search.

```
#include <stdio.h>

// Linear Search Function
int linearSearch(int arr[], int n, int key) {
    for(int i = 0; i < n; i++) {
        if(arr[i] == key)
            return i;  // Return index if found
    }
    return -1;  // Return -1 if not found
}

// Binary Search Function (Array must be sorted)
int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;

    while(low <= high) {
        int mid = (low + high) / 2;

        if(arr[mid] == key)
            return mid;
        else if(arr[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }
    return -1;
```

```
}

int main() {
    int arr[100], n, key, choice, result;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements:\n");
    for(int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    printf("\nChoose Search Method:\n");
    printf("1. Linear Search\n");
    printf("2. Binary Search\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("Enter element to search: ");
    scanf("%d", &key);

    if(choice == 1) {
        result = linearSearch(arr, n, key);
    }
    else if(choice == 2) {
        result = binarySearch(arr, n, key);
    }
```

```
    else {
        printf("Invalid choice!\n");
        return 0;
    }

    if(result != -1)
        printf("Element found at index %d\n", result);
    else
        printf("Element not found\n");

    return 0;
}
```

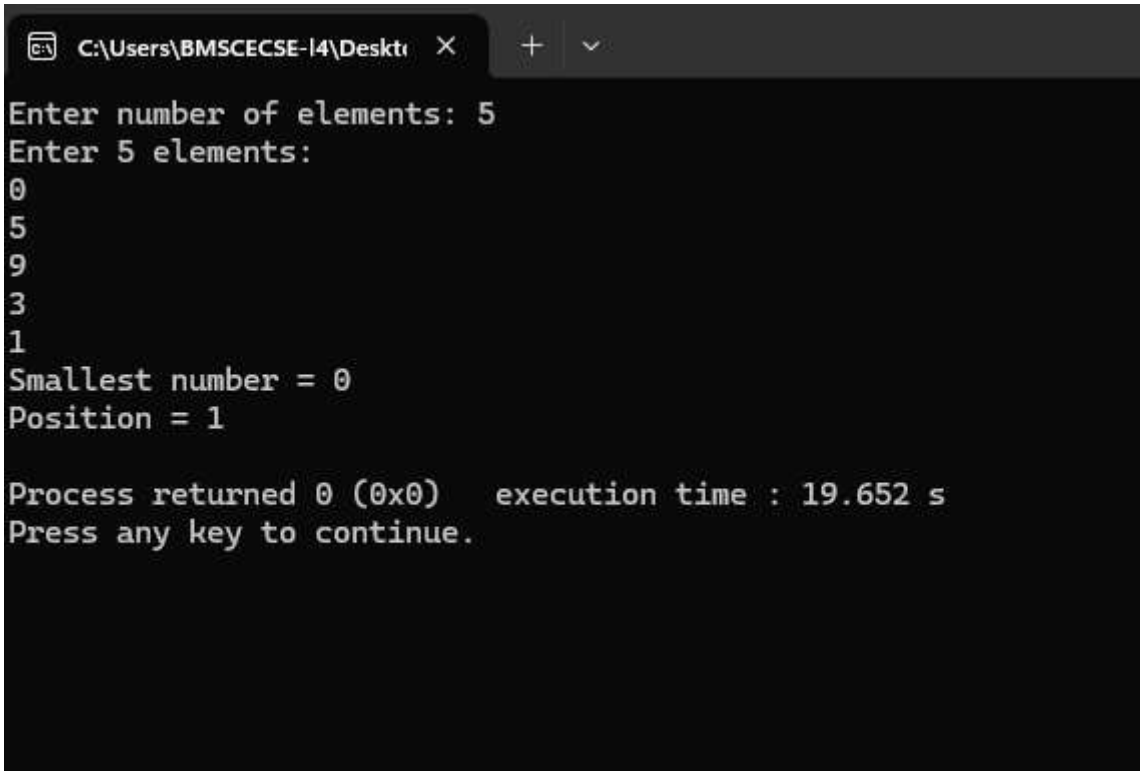


```
Enter number of elements: 5
Enter elements:
12
23
34
45
56

Choose Search Method:
1. Linear Search
2. Binary Search
Enter your choice: 2
Enter element to search: 45
Element found at index 3

Process returned 0 (0x0)   execution time : 16.831 s
Press any key to continue.
```

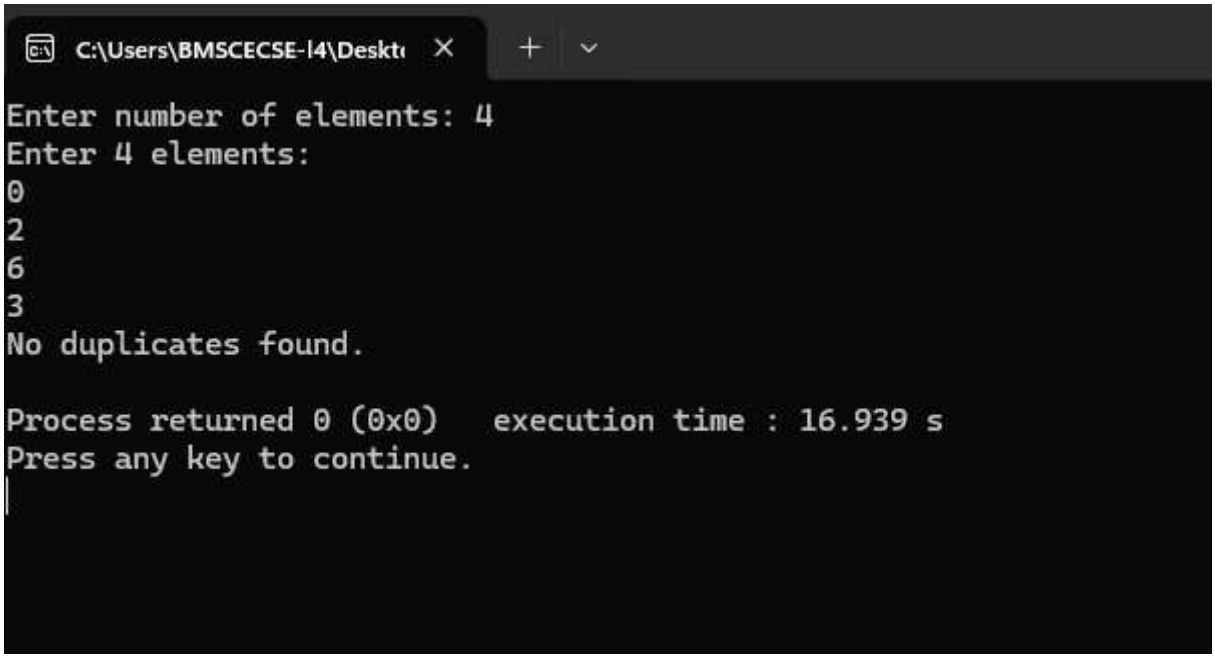
2.Write a C program to print the smallest number and its position in an array of size n.

```
#include <stdio.h>

int main() {
   int arr[100], n;
   int smallest, position;

   printf("Enter number of elements: ");
   scanf("%d", &n);

   printf("Enter %d elements:\n", n);
   for(int i = 0; i < n; i++) {
      scanf("%d", &arr[i]);
   }

   // Assume first element is smallest
   smallest = arr[0];
   position = 0;

   // Find smallest element
   for(int i = 1; i < n; i++) {
      if(arr[i] < smallest) {
         smallest = arr[i];
         position = i;
      }
   }

   printf("Smallest number = %d\n", smallest);
```

```
    printf("Position = %d\n", position + 1);  // +1 for human-friendly position


    return 0;

}
```



3.Write a C program to find if there are duplicates in an array.

```
#include <stdio.h>

int main() {
    int arr[100], n;
    int found = 0;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);
    for(int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    // Check for duplicates
    for(int i = 0; i < n - 1; i++) {
        for(int j = i + 1; j < n; j++) {
            if(arr[i] == arr[j]) {
                printf("Duplicate found: %d\n", arr[i]);
                found = 1;
                break;
            }
        }
        if(found)
            break;
    }

    if(!found)
```

```
    printf("No duplicates found.\n");


  return 0;
}
```